

Connectivity Strategy

Definition

- **Connectivity Strategy** reshapes interactions between firms and customers, transitioning from episodic, reactive interactions to proactive, continuous, and personalized interactions.
- **Aim:** Increase customer satisfaction (Willingness to Pay - WTP) while reducing fulfillment costs simultaneously.

Frameworks, Formulas & Core Ideas

Concept	Description	Formula
Value Created	The total value generated by a firm in the eyes of the customer.	Value Created = WTP - Cost
Competitive Advantage (CA)	Measures the firm's ability to create more value compared to competitors.	$CA = (WTP - Cost)_{You} - (WTP - Cost)_{Competitor}$

How Firms Achieve Competitive Advantage:

Method	Description
Increase WTP	By offering better differentiation and personalized customer satisfaction.
Reduce Costs	Through operational and process efficiencies.

2. Efficiency Frontier

Concept	Description
Efficiency Frontier	Represents the optimal trade-off between WTP and fulfillment costs.
Examples Across the Frontier	High WTP, High Cost: Farmer's Market. Low WTP, Low Cost: Aldi.

Concept	Description
Connected Strategies' Impact	Push the frontier outward by simultaneously increasing WTP and reducing costs.

Firms Redefining the Frontier:

Firm	How They Redefine the Frontier
Blue Apron	Combines convenience with supply chain efficiencies.
Instacart	Reduces friction in grocery shopping through delivery platforms.
Amazon Go	Streamlines in-store experiences with cashier-less technology.

3. Innovation Horizons

Horizon	Description	Example
Horizon 1 (Incremental)	Optimize existing processes and operations.	Scheduling systems to reduce no-shows.
Horizon 2 (Adjacent)	Create new solutions for current customer segments.	AI-assisted sleep apnea diagnostics.
Horizon 3 (Transformational)	Develop radical innovations for new markets or needs.	Transcranial stimulation for sleep disorders.

Connected Strategy Elements

A. Connected Customer Relationships (4 R's Framework)

Step	Description
Recognize	Identify customer needs (explicit or latent).
Request	Help customers articulate their desired product or service.
Respond	Deliver the desired product/service efficiently.
Repeat	Build knowledge to anticipate future needs.

B. Connected Delivery Model

Element	Description
Connection Architecture	Direct delivery, platform-based models (e.g., Uber, Airbnb), or sharing economy models.

Revenue Models	Subscription, pay-per-use, or data monetization models.
Technology Infrastructure	AI, IoT, and automation tools supporting the delivery model.

Key Formulas for Connected Strategy

Metric	Formula
Value Proposition	Value Proposition = WTP – Price
Competitive Advantage	CA = (WTP - Cost)You - (WTP - Cost)Competitor
Optimal Price in Markets	$P^* = WTP_{You} - [(WTP - Cost)_{Competitor}]$ (Set price to offer slightly more net value than competitors.)

Examples of Connected Strategies

Firm	Description
Tesla	Anticipates and resolves customer needs before they arise (e.g., automated tire replacement).
Disney MagicBand	Personalizes customer experiences and reduces operational costs.
Blue Apron	Simplifies meal preparation while optimizing the supply chain for cost efficiency.

Benefits of Connected Strategy

Benefit	How It Is Achieved
Improved Customer Satisfaction (Higher WTP)	Personalized experiences and proactive solutions (e.g., Tesla, Disney MagicBand).
Lower Costs	Efficient resource use and streamlined delivery processes (e.g., Blue Apron, Amazon Go).

Conclusion

- **Connected strategies enable firms to redefine traditional trade-offs by simultaneously:**
 - **Increasing WTP (customer satisfaction).**

- Reducing fulfillment costs.
- **Result: Disruptive competitive advantage that is difficult for competitors to match.**

Five Different Customer Experiences and Levels of Customization

Customer Journey and Connected Customer Experiences

The 4 R's Framework for Connected Experiences

Step	Description
Recognize	Identify the customer need (latent or explicit).
Request	Help the customer articulate the desired product or service.
Respond	Deliver the desired product or service efficiently and effectively.
Repeat	Build knowledge from interactions to continuously improve personalization and customer experience.

Examples of Connected Customer Experiences

Experience Type	Description	Example
Respond-to-Desire	Customers know exactly what they want, and firms focus on fulfilling that need efficiently.	Ordering a specific toner online for a printer and having it delivered within 2 hours.
Curated Offering	Firms help customers explore and choose the best option from a complex set of choices.	Netflix recommending 4 comedies you'll enjoy, instead of listing 10,000 options.
Coach Behavior	Firms nudge customers to take action or make better decisions.	A smart inhaler reminding patients to take their medication and coaching them on proper use.
Automatic Execution	Firms anticipate customer needs and fulfill them without a request.	Printer automatically ordering toner before it runs out, and it's delivered without customer involvement.

Levels of Customization

The **4 Levels of Customization** describe how firms can progressively improve connected customer relationships:

Level	Description	Examples
1. Unified Experience	Creating a seamless experience across all customer touchpoints and episodes.	Disney's MagicBand connects theme park, retail, and hotel experiences for a unified journey.
2. Improved Customization	Using past interactions to better understand individual needs and proactively address them.	A professor tailoring class content based on past student performance and real-time feedback.
3. Meta-Data Insights	Leveraging data across a population of users to enhance product offerings and efficiency.	Netflix creating original content (e.g., "House of Cards") based on insights from viewing trends.
4. Trusted Partner	Addressing customers' deeper needs beyond individual transactions, becoming integral to their long-term goals.	Lynda.com helping users achieve career objectives rather than simply teaching individual skills.

Examples of Innovations in Connected Strategies

1. Smart Inhaler by Adherium

- **What It Does:** Tracks medication adherence, reminds users to take doses, and helps prevent asthma attacks by predicting triggers.
- **Benefits:**
 - Improved health outcomes through reminders and coaching.
 - Feedback for healthcare professionals and pharmaceutical companies to redesign inhalers.

2. Rolls Royce's Aircraft Engine Connectivity

- **What It Does:** Uses sensors to monitor engine conditions and perform predictive maintenance.
- **Key Features:**
 - **"Power-by-the-Hour" Model:** Airlines pay only for usage, aligning incentives between Rolls Royce and customers.

- **Fleet Optimization:** Analyzing entire fleets for efficiency improvements, such as optimal engine washing schedules.

Shifting the Efficiency Frontier

Connected strategies allow firms to **increase willingness-to-pay (WTP)** while reducing costs by:

1. **Streamlining customer journeys.**
2. **Using data-driven insights for predictive and personalized services.**
3. **Building long-term relationships through continuous feedback loops.**

Customer Journey: Key Pain Points and Drivers of WTP

Journey Stage	Pain Points	WTP Drivers
Recognition	Customer unaware of latent needs.	Firms proactively remind or identify needs (e.g., medication reminders).
Request	Overwhelmed by too many options or lack of clear choices.	Curated recommendations tailored to customer preferences.
Ordering & Paying	Complex processes or lack of financing options.	Easy, fast checkout and transparent payment methods.
Receiving	Delays in delivery or the need for customer pickup.	Fast, reliable delivery to the customer's location.
Post-Purchase	Limited flexibility for returns or changes in preferences.	Options for upgrades, returns, or product exchanges.

Key Takeaways on Connected Strategies

1. **Customization Levels:**
 - Firms should focus on building from **basic unification** to **trusted partnerships**, ensuring personalization improves over time.
 - Trust is essential for deeper connections, especially for **automated execution** strategies.
2. **Connected Strategies = Higher WTP + Lower Costs:**
 - Success depends on balancing **efficiency improvements** with **increased customer satisfaction**.

3. Feedback Loops Drive Competitive Advantage:

- Individual-level learning improves service for specific customers.
- Population-level learning enhances products and services for all customers.

Key Elements of Connected Delivery Models

1. Connection Architecture:

- Defines how information flows between the company and the customer.
- Example: A bank acts as a connected producer, offering loans, credit cards, and financial advisory services.

2. Revenue Model:

- Mechanisms through which a company captures the value of its connected services or products.
- Example: Subscription-based model (Dollar Shave Club) or pay-as-you-go model (Car2Go).

3. Technology Infrastructure:

- Technology enabling connectivity and data collection.
- Example: IoT devices, sensors, artificial intelligence.

Connected Strategy and Customer Experiences

Connected Experience	Definition	Example
Respond-to-Desire	Meeting specific and immediate customer needs.	Uber, loan approval (OnDeck, Ernest).
Curated Offering	Providing personalized recommendations based on customer data.	Netflix (content), Wealthfront (personalized portfolios).
Coach Behavior	Helping customers make better decisions through reminders and advice.	Progressive (driving behavior tracking to reduce premiums).
Automatic Execution	Fully automating processes based on customer needs.	Amazon Dash Replenishment, automatic bank savings transfers.

Connection Architectures: Connected Strategy Matrix

Connection Architecture	Respond-to-Desire	Curated Offering	Coach Behavior	Automatic Execution	General Example
Connected Producer	Quick loan approval.	Investment recommendations (banks).	Portfolio rebalancing.	Automatic fund transfers.	Disney MagicBand, Nike.
Connected Retailer	Efficient purchasing/delivery.	Personalized content (Netflix).	Price change alerts.	Automated replenishment.	Amazon, ZipCar.
Connected Market Maker	Simple service booking.	Relevant connections (Expedia).	-	-	LendingTree, OpenTable.
Crowd Orchestrator	Real-time driver availability.	Suggested crowdfunding (Kickstarter).	-	-	Uber, Airbnb, Kickstarter.
P2P Network Creator	Direct transfers.	Active communities (LinkedIn).	-	-	Venmo, Transferwise.

Examples of Innovation Based on Connected Strategy

1. Progressive Customer Segmentation:

- Progressive Insurance uses driving behavior data to personalize premiums.
- Benefit:** Reduces costs and improves customer safety.

2. Car Sharing as a Connected Producer Model:

- Daimler-Benz: Car2Go allows flexible car rentals via an app.
- Benefit:** Maximizes asset utilization and creates ongoing customer interaction.

3. Personalization in Video Games:

- World of Warcraft customizes experiences based on player profiles and skills.
 - Benefit:** Increases willingness to pay and customer loyalty.
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Revenue Models for Connected Products

Case Studies: The Smart Toothbrush

1. Traditional Model (PowerBrush 2050):

- High upfront price: \$600.
- Recurring replacement cost: \$10 per month for brush heads.

2. Connected Model (Smart Connect XL3000):

- Pay-per-use: \$0.10 per minute of brushing.
- Savings-sharing: 50% of dental expense savings with insurers.
- **Advantage:** Continuous customer access and recurring revenue streams.

Principles of Precision Pricing

Principle	Description	Example
Think Value Creation First	Design pricing to maximize value for the entire ecosystem.	Dentists rewarded for healthy teeth.
Trust Your Technology	Charge for outcomes, not products.	Pay for healthy teeth (not toothbrushes).
Think Ecosystem, Not Supply Chain	Include third parties that benefit from the connected product.	Insurers supporting smart toothbrushes.
Pay-As-You-Go	Align pricing with real customer usage.	SaaS software, Car2Go.
Pay with Data	Use customer data as an indirect revenue stream.	Facebook, Google.

Key Insights on Connected Strategies

1. Advantages:

- Enable extreme personalization.
- Increase operational efficiency and reduce costs.
- Facilitate predictable and recurring revenue streams.

2. Risks:

- Privacy concerns: Misuse of personal data.
- Price discrimination: Potential for negative customer perception.
- High upfront costs for technology and data collection.

This matrix and analysis help identify innovation opportunities and evaluate a company's positioning relative to competitors.

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Overview of a Connected Strategy

Component	Description
Connected Customer Relationship	Focused on delivering seamless customer experiences through the 4Rs: Recognize, Request, Respond, Repeat.
Connected Delivery Model	Involves Connection Architecture, Revenue Models, and Technology Infrastructure to enable the connected strategy effectively.

The STAR Framework

STAR Element	Description	Example
Sense	Detect user needs or events.	A sensor identifies the coffee supply is running low.
Transmit	Send data to the relevant system.	Data is sent to the cloud for processing.
Analyze	Process and interpret the data to create insights.	The system determines whether a coffee refill is required.
React	Respond to insights by initiating actions.	Automatically place a coffee order or notify the user.

Deconstructing the Coffee-Making Example

The following table illustrates how the STAR framework applies to the coffee-making connected strategy across the 4Rs and connected delivery model.

STAR Dimension	Recognize	Request	Respond	Repeat	Connection Architecture	Revenue Model
Sense	Detect low coffee supply.	Identify user's coffee preference.	Detect user arrival at the door.	Collect user satisfaction metrics.	Identify ecosystem needs.	Monitor coffee machine performance.
Transmit	Send stock level to cloud.	Send order to retailer.	Notify coffee machine to start brewing.	Share feedback with system.	Share preferences with partners.	Transmit maintenance needs to provider.
Analyze	Check if stock is below threshold.	Evaluate available coffee options.	Confirm user identity.	Assess user delight.	Optimize partner connections.	Decide when to repair or replace machine.
React	Trigger a refill request.	Place coffee order.	Start brewing coffee.	Adjust future orders.	Implement system improvements.	Dispatch technician or replacement machine.

Tool 1: Classification Trees

What are Classification Trees?

Classification trees help explore and categorize all possible solutions for a specific function, e.g., identifying a person.

Function	Category	Sub-Category	Examples
Recognize a person	Human-Based Solutions	Manual	Security guard.
	Automated Solutions (User Action)	Enter PIN or swipe card	Gym PIN codes or keycards in hospitals.

	Automated Solutions (No Action)	NFC/Bluetooth proximity sensors	Car keys, smartphone geofencing.
		Biometric recognition	Face or fingerprint recognition.

Tool 2: Selection Tables

Selection Table for Recognizing a Person at a Door

This table compares various technologies based on performance, cost, and application:

Technology	Convenience	Safety	Cost	Applications	Comments
Human at Door	++	++	--	Hotels	Reliable but costly.
Enter PIN	-	+	++	Gyms	Easy to implement, low cost.
NFC Sensor Proximity	++	+	+	Car keys	Fast and widely used.
2D Face Recognition	++	+	+	Hotels, Phones	Convenient but moderate cost.
Fingerprint Recognition	-	++	+	Smartphones	Secure but less convenient.

Bottom-Up Innovation

Emerging technologies can unlock new connected strategies by enabling previously infeasible solutions. Here's an example of how technologies bubble up through the stack.

Technology	Innovation Enabler	Example Application
Voice Recognition	Improved data processing and machine learning algorithms.	Virtual assistants like Siri or Google Assistant.
3D Printing	Customization at low cost.	Personalized medicine dosages or custom hardware.
Blockchain	Secure and transparent data transmission.	Tracking supply chains or enabling cryptocurrency.
5G and Bluetooth LE	Faster data transmission with lower energy consumption.	Real-time updates for IoT devices, like connected cars.

Example Technology Stack

For a Smart Coffee Maker, the technology stack can be deconstructed as follows:

Layer	Example Technologies	Functionality
Sensing	IoT sensors, weight scales	Detect low coffee supply.
Transmission	Wi-Fi, Bluetooth, 5G	Send data to the cloud for processing.
Analysis	Cloud computing, machine learning	Identify patterns in coffee consumption.
Reaction	IoT actuators, drones	Place order and deliver coffee pods.

Technological Innovations by STAR Framework

The STAR framework can also be used to track emerging technological advances:

STAR Element	Technological Innovations	Impact
Sense	IoT sensors, wearable devices, augmented reality	Improve ability to detect customer needs.
Transmit	5G, Bluetooth LE, LiFi	Enable fast and energy-efficient communication.
Analyze	Machine learning, quantum computing, advanced cloud storage	Enhance predictive capabilities and insights generation.
React	Autonomous drones, AI-driven personalization, advanced robotics	Deliver faster and more accurate responses to user needs.

Key Takeaways

1. **Use the STAR Framework: Deconstruct strategies by mapping sub-functions across Sense, Transmit, Analyze, and React for each customer interaction (Recognize, Request, Respond, Repeat).**
2. **Leverage Classification Trees and Selection Tables:**
 - Classification trees help brainstorm potential solutions.
 - Selection tables evaluate and compare options based on relevant criteria.

- 3. Keep Track of Bottom-Up Innovation:** Emerging technologies can disrupt traditional approaches, enabling new business opportunities.
- 4. Balance Cost and Efficiency:** Use connected technologies to streamline processes, reduce costs, and improve customer experiences.